Science 9 Mid-Term Test

Energy and Change & Investigating Scientifically

Name: _____

Outcome	Pointer	Questions	Achieved?
IS2	• Recognise which variable is the focus of the investigation	1, 2, 4	
IS 3	• Organises numerical data in to tables	3, 5	
E&C2	• Describes how energy can be used in daily life • Describe ways energy is transferred	1, 6, 7	
E&C3	• Understands some types of energy transfer and relates types of energy transfer to the state of matter of the conducting materials	1, 6, 7, 8, 9, 10	
E&C4	• Compares different methods of conduction and insulation • Describes how colour and texture affect an object's ability to absorb heat • Identifies the ways in which heat energy interacts with different substances, causing contraction and expansion	1, 2, 3, 4, 5, 6	
E&C5	• Explains, at the particle level, energy changes that occur during change of states.	11	

Part One: **Investigating Scientifically**

Introduction

At the front of the room there is an experiment set up. There are four aluminium cans, one of which is plain, while the others are covered with various materials. The teacher fills each can with boiling water from a kettle, and records the temperature of the water in the cans. Measurements will be taken every minute for ten minutes.

1. **AIM:** What do you think this experiment is designed to investigate?

_____

_____

_____

2. **MATERIALS:** What equipment is being used in this investigation?

_____

_____

_____

_____

_____

3. **RESULTS:** Design and construct a table to record the results of this experiment

4. **METHOD:** In this experiment, what is the dependent variable? (i.e. What is the variable we are measuring?

_____

5. **ANALYSIS:** How will you assess the effectiveness of the different insulative coverings?

_____

_____

_____

Part One: **Energy and Change**



Try to use the following scientific words when you answer the questions below. If you can't then use your own words:

Atoms	**Energy**	**Heat**	**Conductor**	**Insulator**
Reflect	**Absorb**	**Conduction**	**Convection**	**Radiation**
Thermometer	**Expand**	**Contract**	**Vibrate**	**Float**

1. List two materials you used to make your esky and say how each material helps keep the drinks inside cold. (Use words from the list)

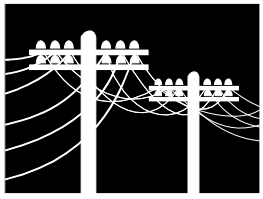
   a. Material 1:

      How it helped keep the drinks cold:

   b. Material 2:

      How it helped keep the drinks cold:

2. Why do railway lines have gaps, power lines hang down and concrete driveways have rubber in gaps between sections? (Think about what happens to them on hot or cold days, and remember to try and use words from the list above)



3. White is the most common colour for cars sold in the northwest followed by silver then red then blue. How come? (Use words from the list)



4. A quick way to cook tinned food when camping is to put the tin directly in the fire. What should you do to make this safe? Why? (Try to use words from the list)



5. Hot air balloons only contain hot air. How can they lift a heavy basket with two people inside. (Remember to use the words from the list)



6. List three design ideas to keep a Broome house as cool as possible without using electricity (air conditioners or fans)? Explain how each idea helps.



a. Idea 1:

How it keeps the house cool:

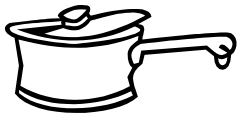
b. Idea 2:

How it keeps the house cool:

c. Idea 3:

How it keeps the house cool:

7. Saucepans and frypans always have metal bases and usually have plastic or wooden handles. Why? (Try to use words from the list)



8. Name three conductors and three insulators. (You will need to use your own words)

   a. Conductors:

   b. Insulators

9. Explain how heat moves through a solid. What is this type of heat transfer called?

10. Explain (or draw a diagram to show) how heat moves through a liquid or gas. What is this type of heat transfer called?

11. What happens to the particles in a block of ice as it melts?

Harder questions for experts

10. Water in the pool feels cooler even though the water is the same temperature as the air. Why?

11. Air conditioners in houses and classrooms usually have the vents that the cold air comes out of either high up in the wall or in the roof. Why are the vents always placed high up in a room and never near the floor?

12. If someone were to throw boiling hot water on you, would you prefer a cupful of water or a bucketful of water? Why? (Try to use words from the list)



End of Assessment

Check your answers if you finished early. Good Luck.